

Recent Advances in C(sp³)–C(sp³) Cross-Coupling Chemistry: A Dominant Performance of Nickel Catalysts

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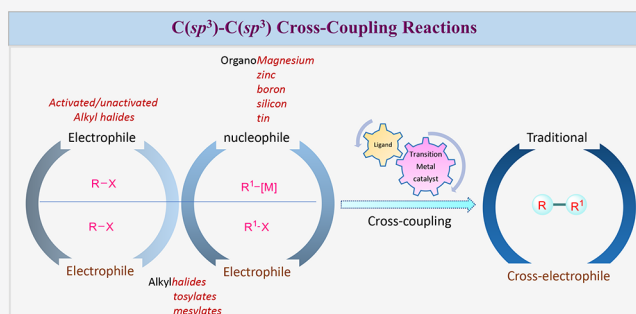
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ABSTRACT: The formation of a new C–C bond at sp³-hybridized centers is highly desirable, as it opens up a unique, unexplored chemical space. The revolutionary discoveries in this field meet the longstanding challenge of forming stereocontrolled C–C bonds and provide rapid access to C(sp³)-rich drug molecules. The new and improved catalytic systems have enabled the creative design of challenging transformations. The latest advancements in nickel catalysis are particularly attractive and have shown excellent catalytic performance in this field. The progress that has been made toward C(sp³)–C(sp³) bond formation (traditional and cross-electrophile couplings) in the past decade is highlighted in this review.



INTRODUCTION

Transition-metal-catalyzed reactions for the construction of alkyl–alkyl bonds are the most versatile types and have become prospects for the next revolution in organic synthesis.¹ An sp³-hybridized carbon has a three-dimensional spread; due to this, the hydrophilicity of the compound is greatly enhanced.² Therefore, C(sp³)–C(sp³) cross-coupling reactions have attracted interest in exploring the chemical space by introducing more asymmetric sp³-hybridized carbons into drug candidates. An excellent review on the applications of C(sp³)–C(sp³) cross-coupling reactions in the synthesis of natural products and bioactive compounds was published by Kirschning et al.³

Despite the usefulness of classical S_N1 and S_N2 reactions in organic synthesis, limitations in scope and stereocontrol cause difficulties in nucleophilic substitution reactions of alkyl electrophiles. An intriguing way to address these challenges makes use of transitional-metal-catalyzed cross-coupling reactions. This concept has displayed an impressive scope of reaction classes toward new carbon–carbon bond formation. The reliability and flexibility of cross-coupling reactions have enabled the widespread use of these reactions in all areas of synthetic chemistry.⁴ Even with the development of cross-coupling reactions in organic synthesis, most of the bond-forming reactions are limited to C(sp³)–C(sp²) and C(sp²)–C(sp²) couplings.⁵ Due to the difficulties arising from the sluggish oxidative addition of C(sp³) electrophiles and the β-hydride elimination of alkyl–metallic intermediates, metal-catalyzed C(sp³)–C(sp³) cross-coupling reactions remained unexplored until the beginning of this century.⁶ Several decades after the development of cross-couplings of unsaturated compounds, Knochel started a new era of TM-catalyzed cross-coupling

between two C(sp³) centers in 1996.⁷ The award of 2010 Nobel prize in chemistry to Heck, Negishi, and Suzuki for the discovery of palladium-catalyzed cross-couplings in organic synthesis has inspired chemists to create flourishing new dimensions in cross-coupling reactions.⁸ The massive advancements in alkyl–alkyl cross-coupling chemistry are likely to expand the scope and suppress formation of the homocoupling byproducts; thus, there are enormous opportunities available for inventions around stereoselective cross-couplings. Fu and co-workers have made significant advances in this area, and their intense research effort expanded the breadth of these reactions to the most advanced double-enantioconvergent coupling reactions.⁹ Fu also published a review on transition metal-catalyzed traditional alkyl–alkyl cross-coupling reactions, covering the literature until the end of 2016.¹⁰ Recent reports by Gong,¹¹ MacMillan,¹² and Jarvo¹³ made use of secondary electrophiles and have provided new avenues for developing enantioselective carbon–carbon bond formation reactions. In view of the interest in C(sp³)–C(sp³) cross coupling chemistry, new developments during the period from 2011 to 2021 are the focus here. Investigating the suitable metal catalyst, ligand, and coupling partners are the core elements in fine-tuning of the cross-coupling reactions.

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Nickel (an earth-abundant metal) catalysts play a dominant role in these reactions.^{14–16} The inexpensive nickel catalysts have several advantages over palladium catalysts for cross-coupling reactions (Figure 1). Electrophile activation is an important

28 Ni 58.693 Nickel [Ar] 3d⁸ 4s²	Inexpensive
	Ni(I), Ni(III) stable oxidation states
	facile oxidative addition to C(sp ³)-electrophiles
	Slower β -H elimination

Figure 1. Unique features of nickel catalysts in comparison to palladium and copper.²⁰

aspect for the successful outcome of cross-coupling reactions. Due to the balanced open-shell electronic configuration¹⁷ and the lower electronegativity,¹⁸ the nickel catalysts are able to facilitate efficient oxidative addition with the C(sp³) electrophiles. In addition, the slower rate of undesired β -H elimination of alkyl-[Ni] intermediates make these catalysts more suitable for reactions employing alkyl halides and pseudohalides.¹⁹ On account of this, the C(sp³)-coupling partners that decompose via β -hydride elimination with other catalysts are more compatible with the nickel catalysts. The Diao research group has conducted an extensive mechanistic investigation to better understand the reactivity and rationale of the Ni-catalyzed cross-coupling reactions.^{17,20} In view of their importance, in addition to Ni-catalyzed protocols, Fe- and Cu-catalyzed methods are also reviewed here.

A decade of development in this challenging area has provided a flexible toolkit for the new carbon–carbon bond formation. The useful toolbox for the C(sp³)-C(sp³) cross-coupling reactions depicted in Figure 2 may guide the discovery, optimization, and creation of efficient catalytic systems to further expand the scope of this thriving area.

The traditional cross-coupling reactions (Scheme 1a) and the cross-electrophile coupling reactions (Scheme 1b) are mainly highlighted in this review. Except where noted, direct C(sp³)-H alkylation reactions²¹ and cross-dehydrogenative couplings²² are not covered here. The hope is that the transformations and the mechanisms presented herein will provoke further advancements in creating new chemical bonds.

TRADITIONAL CROSS-COUPLING REACTIONS

Cross-coupling reactions between electrophiles, such as organic halides and nucleophilic organometallic reagents, are the most practical and valuable methods for the formation of new C–C bonds in organic synthesis. Earlier studies of such reactions were mostly used palladium catalysts to construct a bond between sp²-hybridized carbons.²³ However, in recent years tremendous advancements have been made with nickel-based catalysts to create a bond between sp³-hybridized carbons.

Mechanism. In general, nickel-catalyzed cross-couplings are based on the classical Ni(0)/Ni(II) catalytic cycle. The proposed mechanism begins with oxidative addition of Ni(0) complex **A** to an alkyl halide, followed by a transmetalation and reductive elimination sequence (Scheme 2a). However, with redox-active ligands (terpyridine, bipyridine, phenanthroline, and analogues) the reactions are likely to involve Ni(I)/Ni(III)

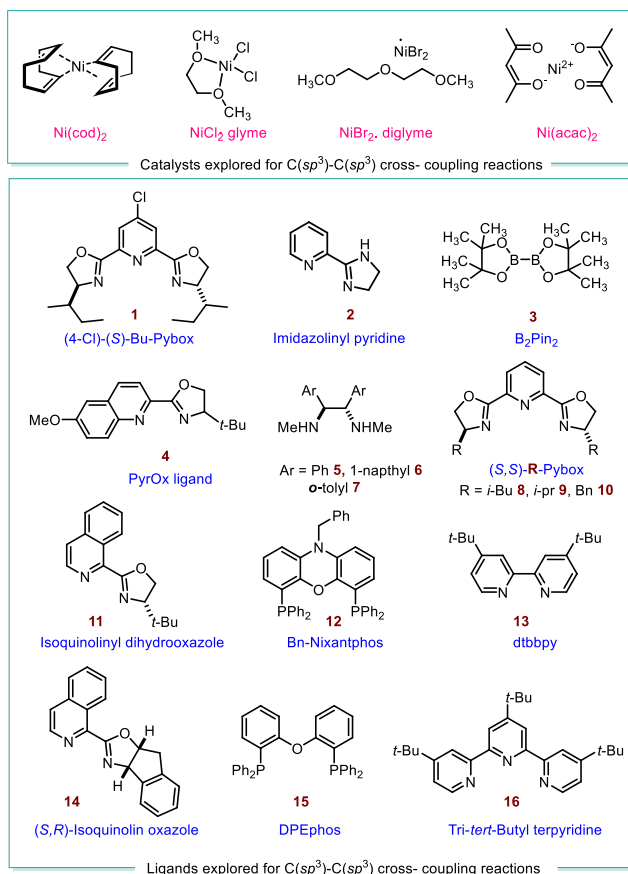
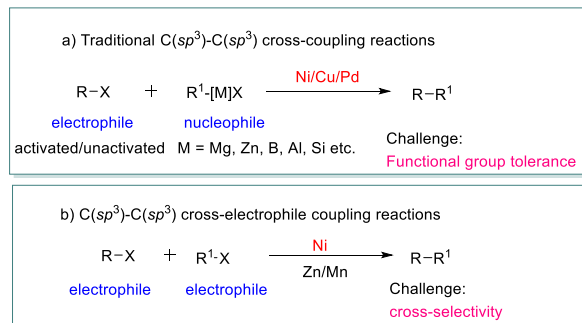
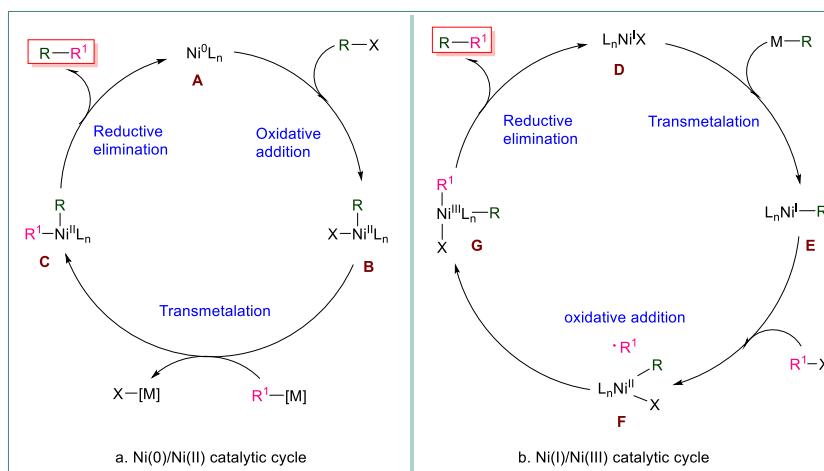


Figure 2. Useful toolbox for C(sp³)-C(sp³) cross-coupling reactions.

Scheme 1. Transition-Metal-Catalyzed C(sp³)-C(sp³) Cross-Coupling Strategies



catalytic cycles.²⁴ The easy accessibility of Ni(I) and Ni(III) oxidation states enables the different modes of reaction mechanisms. The Ni(I)/Ni(III) catalytic cycle proposed by Fu proceeds through a radical oxidative addition pathway (Scheme 2b).^{24c,d} Transmetalation of the organometallic nucleophile to the Ni(I) complex **D** occurs before the oxidative addition step. Capture of a radical by the Ni(II) intermediate **F** or oxidative addition to the Ni(I) intermediate **E** generates the Ni(III) species **G**, which then undergoes reductive elimination to furnish the coupled product. A series of extensive mechanistic investigations by Diao,^{17,20} Martin^{24e} and Vici^{24a} groups have provided evidence for the single-electron-transfer (SET) processes. However, the first isolation and characterization of Ni(III) alkyl species was demonstrated by the Tilley group.^{24f} The possible pathways for the C(sp³)-C(sp³) cross-coupling

Scheme 2. Possible Mechanisms for Nickel-Catalyzed C(sp³)–C(sp³) Cross-Coupling

reactions with nickel catalysts are illustrated in Scheme 2. Additionally, due to the focus on nickel dominance in this review, mechanistic discussions regarding iron^{24g} and copper-catalyzed^{24h} cross-coupling reactions will not be evaluated.

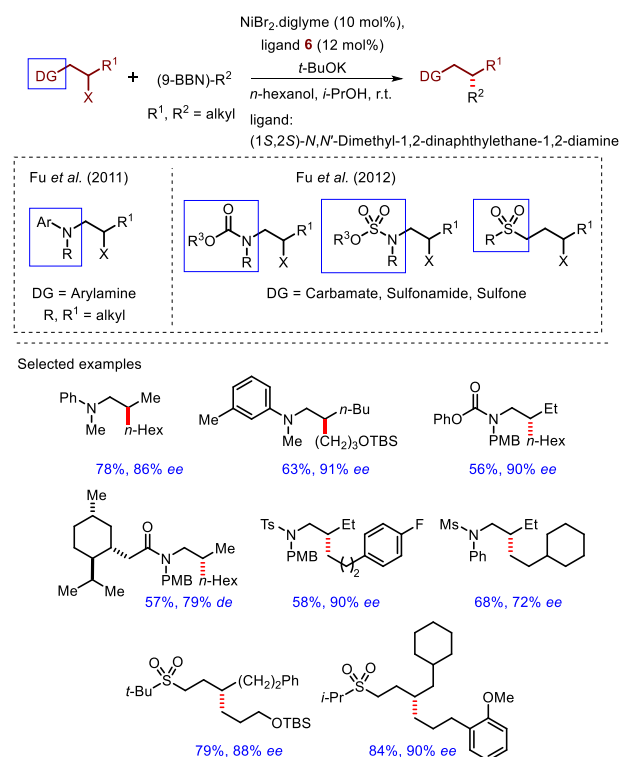
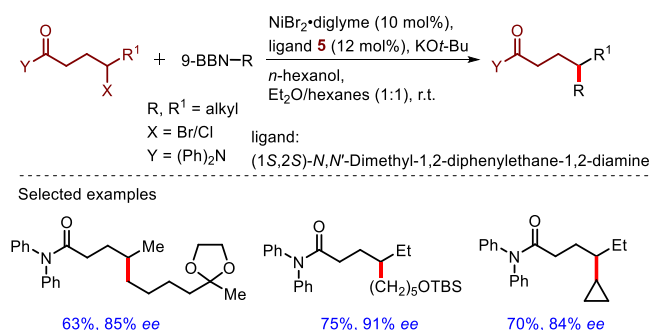
C(sp³)–C(sp³) Cross-Coupling Reactions of Unactivated Alkyl Electrophiles. The use of unactivated alkyl electrophiles as some of the coupling partners is highly challenging due to the fact that the slower oxidative addition of metal complexes to the alkyl halides/sulfonates leading to thermodynamically favored intramolecular β -hydride elimination results in the formation of undesired olefinic byproducts.²⁵ Due to the concern about this difficulty, the cross-coupling reactions of unactivated alkyl electrophiles were not investigated in the early 21st century. There exist only three examples of these kind of processes before 2011.^{16a–c} However, in recent years this area has experienced significant progress driven by the ready availability of alkyl halides. The rate of progress in this field has increased significantly after Fu's discovery of Pd-catalyzed Suzuki coupling of unactivated alkyl bromides.²⁶ Reactions in which one of the electrophiles is initially turned into an organometallic reagent in the presence of a reducing metal, followed by the classical cross-coupling reactions, were also covered here.

In 2011, Fu and co-workers discovered an efficient cross-coupling reaction between γ -halo carbonyl compounds and alkylborane reagents (Scheme 3).²⁷ The optimized conditions revealed that, in the presence of 10 mol % of NiBr₂·diglyme, 12 mol % of the ligand, 1.8 equiv of 1-hexanol, and Et₂O/hexanes (1/1) as solvent, the reaction of γ -chloro/-bromo carbonyl

compounds with alkyl-(9-BBN) derivatives produced enantio-rich alkylated compounds. Notably, a secondary alkylborane (cyclopropane) reagent could also be introduced as an efficient substrate in this catalytic system.

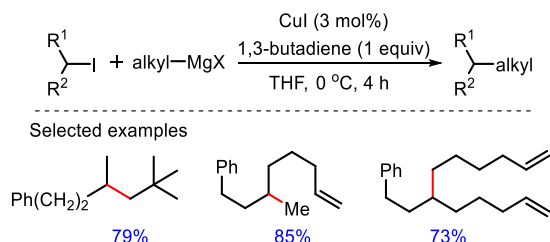
In the same year, the Fu laboratory also explored the application of *nitrogen*-directed groups in the context of stereoconvergent alkyl–alkyl cross-couplings.²⁸ The optimized protocol allows the enantioselective cross-couplings of unactivated secondary alkyl chlorides bearing a pendant amine, under a chiral nickel catalytic system in the presence of alkylboranes as coupling partners (Scheme 4). One year later, the same authors published an extension of this directing-group-assisted asymmetric cross-coupling reaction, enabling the creation of carbamates, sulfonamides, and sulfones.²⁹

Scheme 4. Directing-Group-Assisted Enantioselective Suzuki Cross-Coupling of Unactivated Secondary Alkyl Halides

Scheme 3. Cross-Coupling of γ -Haloamides with Alkylboranes

The classical Ullmann chemistry served well for a large number of C–C bond forming reactions for over a century. However, copper-assisted cross-coupling reactions³⁰ have been overlooked after the advent of palladium catalysis. Kambe and co-workers succeeded in coupling Grignard reagents to secondary unactivated alkyl halides using a catalytic amount of CuI (Scheme 5).³¹ The reaction was assumed to proceed

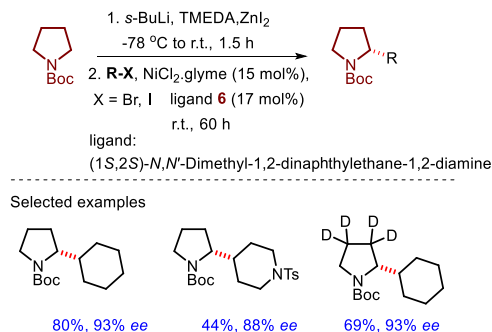
Scheme 5. Cross-Coupling of Unactivated Secondary Alkyl Iodides with Alkyl Grignard Reagents



through the formation of magnesio cuprates, which then react with the alkyl halides to furnish the desired product. The use of 1,3-butadiene as an additive is critical to the success of this reaction. It is believed that 1,3-butadiene stabilizes the cuprate intermediate by restricting β -hydride elimination and other side reactions (Scheme 5).

The Fu group also disclosed the enantioconvergent alkyl–alkyl cross couplings of *N*-Boc-pyrrolidine with secondary alkyls (Scheme 6).³² This method allows the synthesis of 2-

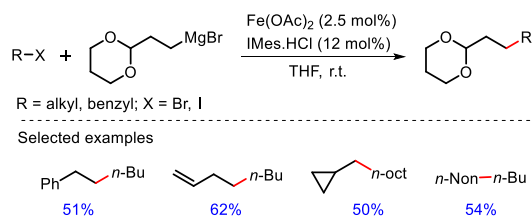
Scheme 6. Enantioconvergent Cross-Coupling of *N*-Boc-pyrrolidine with Secondary Alkyl Halides



alkylpyrrolidines, which are common constituents of many alkaloids. A variety of cyclic and acyclic alkyl iodides/bromides were successfully cross-coupled with α -zincated *N*-Boc-pyrrolidine in the presence of NiCl₂·glyme and a chiral 1,2-diamine ligand with good enantioselectivity (Scheme 6). In contrast to the studies reported by Kumada and co-workers,³³ the product stereochemistry is mainly controlled by the stereochemistry of the chiral nickel catalyst.

In 2013, Cárdenas and co-workers claimed the first iron-catalyzed cross-coupling between unactivated alkyl halides and organomagnesium reagents (Scheme 7).³⁴ The authors reported a Kumada type cross-coupling with a catalytic system containing 2.5 mol % of Fe(OAc)₂ and 6 mol % of dimesitylimidazolium chloride as the ligand. The reaction runs in THF at 50–60 °C and requires an excess of alkylmagnesium bromide (1.7 equiv) for better yields. However, lower yields are observed with alkyl bromides and tosylates and there is no reaction with alkyl chlorides. The proposed mechanism involves a transmetalation

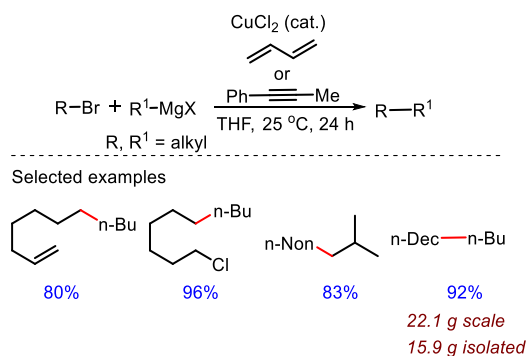
Scheme 7. Cross-Coupling of Alkyl Halides with Alkylmagnesium Reagents



with a Grignard reagent followed by oxidative addition and then reductive elimination.

In a continuation of their previous studies, Kambe and co-workers reported the cross-coupling of unactivated alkyl halides with Grignard reagents with a catalytic system comprised of CuCl₂ and a butadiene or phenylpropyne additive (Scheme 8).³⁵ The authors were convinced that the use of hydrocarbon

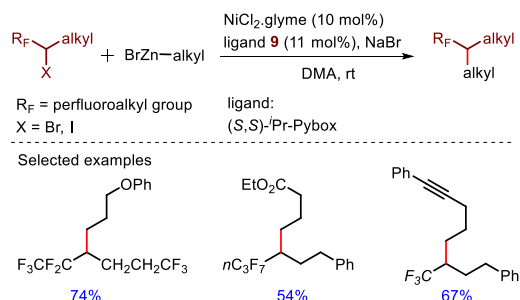
Scheme 8. Cross-Coupling of Unactivated Alkyl Halides with Grignard Reagents



additives prevents the autocatalytic decomposition because of the formation of Cu–alkyne complexes, which suppress the aggregation of the copper catalyst. The practical utility of this reaction was demonstrated by handling the reaction on a 22.0 g scale.

Fu et al. reported in 2015 the alkyl–alkyl Negishi cross-coupling of fluorinated secondary electrophiles catalyzed by NiCl₂·glyme (Scheme 9).³⁶ In this reaction manifold, 10 mol % of catalyst, 11 mol % of ligand 9, and 1.0 equiv of NaBr in DMA generated the desired perfluoroalkylated compounds in good yields. Further studies revealed that the level of fluorination of the alkyl halide greatly influences the reaction rate. A further application of the optimized conditions displayed that a

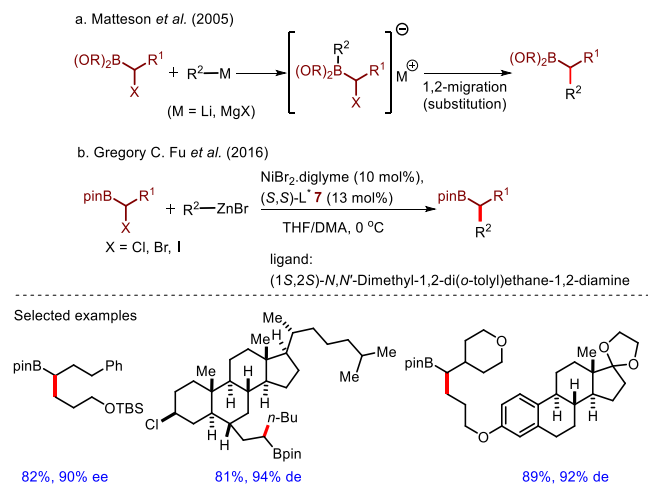
Scheme 9. Negishi Coupling of Fluorinated Secondary Alkyl Halides



fluorinated alkyl iodide could also be employed and could give the product in good yield.

Inspired by the art of enantioselective transition-metal-catalyzed $C(sp^3)-C(sp^3)$ couplings of racemic alkyl electrophiles and carbon nucleophiles, Fu and co-workers initiated a research program with the aim using a different electrophile that bears α -haloboronates (Scheme 10b).³⁷ This process displays an

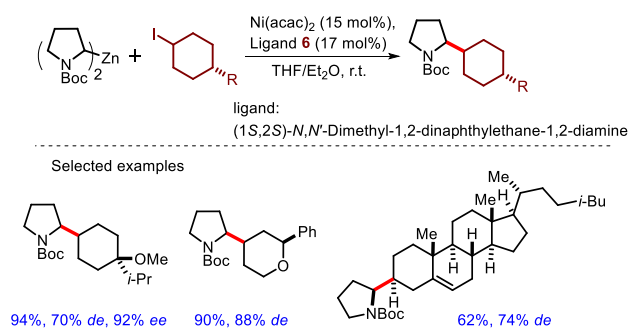
Scheme 10. Asymmetric Negishi Type Cross-Coupling of α -Haloboronates



improved functional group tolerance and has an advantage over the process reported previously by Matteson (Scheme 10a), since the coupling is carried out by organozinc nucleophiles generated under mild conditions and uses a chiral catalyst instead of a stoichiometric chiral reagent. The present protocol allows easy access to a range of alkylboronate esters, thereby offering the extra dimension of further functionalization to furnish a variety of enantioriched organic frameworks. The racemic α -haloboronates are converted into enantioriched alkylboronates by the chiral nickel diamine catalyst via a radical intermediate generated from homolytic cleavage of the C–Cl bond.

A recent major advancement in cross-coupling chemistry is Fu's development of alkyl–alkyl cross coupling of *N*-Boc-pyrrolidine with secondary alkyl halides, which showcased a doubly stereoconvergent alkyl–alkyl bond formation.³⁸ The authors noted that the coupling of 2-zincated *N*-Boc-pyrrolidine with a 4-substituted cyclohexyl halide was successful and provided excellent stereoselectivity under the previously described conditions (Scheme 11). Their studies revealed that

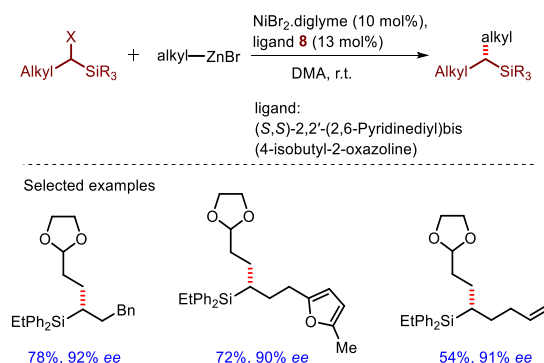
Scheme 11. Doubly Stereoconvergent Cross-Coupling of Cyclic Alkyl Halides



the chiral catalyst which controls the stereochemistry of the carbon belongs to the pyrrolidine ring, as reported previously, and the substrate controls the stereochemistry at the other carbon atom of the newly formed carbon–carbon bond. This is an early notable example of an alkyl–alkyl coupling reaction with a good control of vicinal stereocenters.

In 2019, the same group was able to expand their previous methodology to the use of α -halosilanes as electrophiles (Scheme 12).³⁹ It was found that α -bromo/ α -iodosilanes

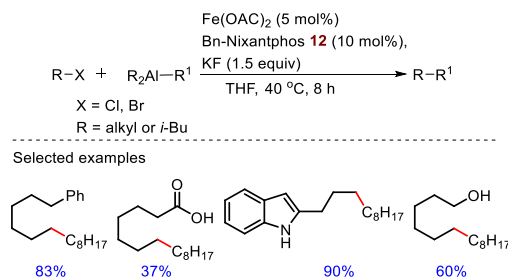
Scheme 12. Enantioconvergent Negishi Coupling of Racemic α -Halosilanes



reacted with alkylzinc bromides in the presence of NiBr₂·diglyme catalyst and a chiral pybox ligand to give chiral organosilanes with excellent stereoselectivity. The reaction was envisaged to proceed through the formation of an alkyl radical and alkylnickel(II) intermediates that can combine to form [L*Ni^{III}R(R¹)]Br followed by reductive elimination leading to the coupling product. A control experiment suggested that the chirality of the catalyst rather than the kinetic resolution plays a dominant role in determining the ee of the product.

In 2019, Masaharu and co-workers revealed another Fe-catalyzed alkyl–alkyl cross-coupling of unactivated alkyl halides with alkylaluminum reagents (Scheme 13).⁴⁰ The catalytic

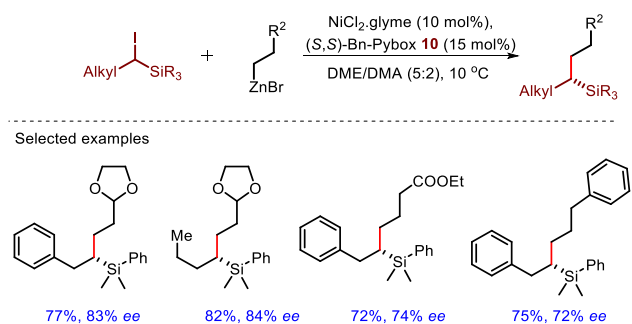
Scheme 13. Cross-Coupling of Unactivated Alkyl Halides with Organoaluminum Reagents



system is composed of Fe(OAc)₂, the Bn-Nixantphos ligand, and KF as an additive. The fluoride anion facilitates the transmetalation by generating active aluminate species. The mild reaction conditions displayed a good chemoselectivity.

Simultaneously, Oestreich *et al.* communicated a similar coupling reaction of α -iodoalkylsilanes and alkylzinc reagents.⁴¹ In this study, racemic α -iodoalkylsilanes were treated with alkylzinc bromides in the presence of NiCl₂·glyme and (S,S)-Bn-Pybox, affording enantioriched $C(sp^3)-C(sp^3)$ coupling products (Scheme 14). In contrast to Fu's report, the reaction did not work well with an α -bromoalkylsilane and the reason was

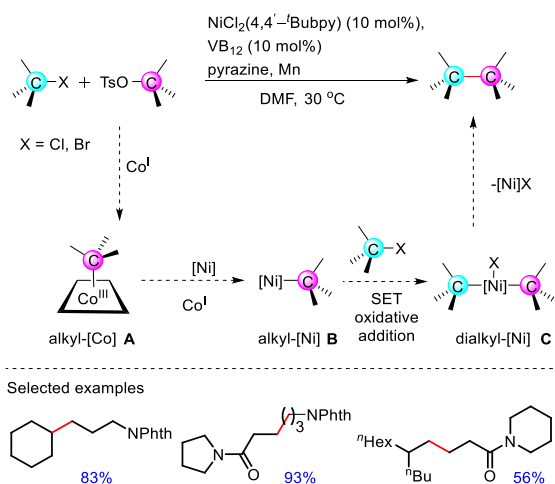
Scheme 14. Enantioselective Cross-Coupling of Racemic α -Silylated Alkyl Iodides with Alkylzinc Reagents



not explained. The authors proposed that the reaction proceeds through the formation of α -alkylsilane radical and alkylnickel(II) intermediates. The subsequent reaction of these two intermediates followed by reductive elimination affords the desired product. The α -silicon effect was clearly determined by replacing the silyl group with a methyl group, as no cross-coupling product was observed.

More recently Komeyama et al. developed a $C(sp^3)-C(sp^3)$ cross-coupling without the aid of alkyl-metallic reagents.⁴² The proposed $C(sp^3)-C(sp^3)$ cross-coupling involves the reaction between alkyl tosylates and alkyl halides using a hybrid catalytic system made with nickel and cobalt catalysts (Scheme 15).

Scheme 15. Cross-Coupling of Alkyl Halides with Alkyl Tosylates

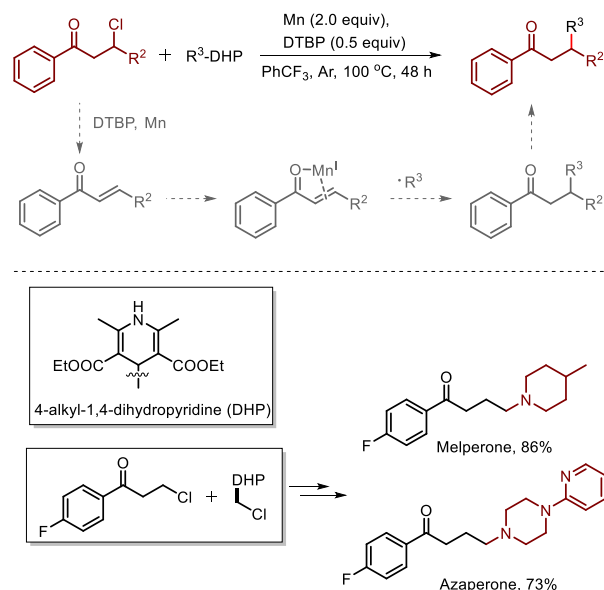


Here, alkyl tosylates were successfully utilized for $C(sp^3)-C(sp^3)$ couplings with their newly developed Ni/Co catalytic system and offered a novel synthetic approach for a variety of $C(sp^3)-C(sp^3)$ linkages. The method used planar cobalt species, obtained from the reduction of vitamin B₁₂, which underwent an initial S_N2 -type oxidative addition of alkyl tosylates to furnish the alkyl-[Co] intermediate A. This intermediate evolves through a transmetalation with low-valent nickel to give the alkyl-[Ni] intermediate B. In this case the alkyl halides could add to the alkyl-[Ni] intermediate via an SET-type oxidative addition (SET-OA) instead of the cobalt intermediate to afford the dialkyl-[Ni] species C, thereby delivering a cross-coupling product through reductive elimination.

In 2019, Zhao et al. reported a manganese-induced alkyl-alkyl cross-coupling of β -chloroketones with alkyl-DHPs

(dihydropyridine) (Scheme 16).⁴³ The use of alkyl-DHPs has the great advantage of their being easily cleaved at the 4-position

Scheme 16. Cross-Coupling of β -Chloropropiophenones and Alkyl-DHPs

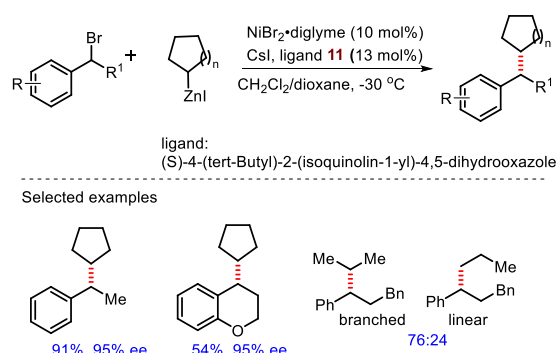


of $C-C$ to generate alkyl radicals, and these compounds can be easily prepared from aldehydes in a single step.⁴⁴ The treatment of β -chloropropiophenone with various alkyl-DHPs in the presence of DTBP (2-(*tert*-butylperoxy)-2-methylpropane) results in dehydrochlorination. The Mn complex then coordinates with a vinyl ketone followed by radical addition and subsequent protonation to furnish the desired coupling products. More interestingly, the method was also applied to the synthesis of the bioactive compounds melperone (well-known drug for schizophrenia) and azaperone.

$C(sp^3)-C(sp^3)$ Cross-Coupling Reactions of Activated Alkyl Electrophiles. The earlier developments of catalytic asymmetric alkyl-alkyl cross-couplings revolved around the use of activated alkyl halides as electrophilic partners. A competitive isomerization of the nucleophile is the prevailing side reaction in the cross-coupling of activated electrophiles.⁴⁵

In 2012, Fu and co-workers developed an enantioselective cross-coupling of racemic benzyl bromides with achiral cycloalkylzinc nucleophiles (Scheme 17).⁴⁶ The reaction was performed using 10% $NiBr_2$ ·glyme and a bidentate isoquinoline

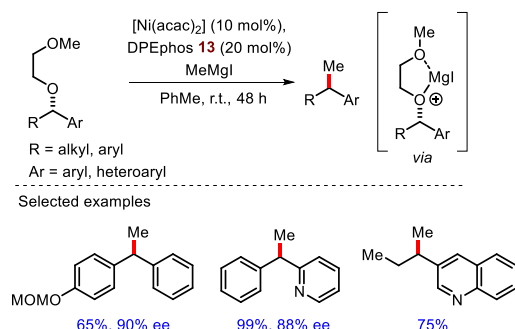
Scheme 17. Enantioselective Cross-Coupling of Racemic Alkyl Halides with Achiral Alkylzinc Reagents



line–oxazoline ligand. The formation of a significant amount of branched product was observed in the case of a primary alkylmetallic nucleophile due to isomerization. The racemic benzyl bromides are transformed into the enantioriched cross-coupled products.

Installation of a methyl unit onto specific organic frameworks results in an increase in binding affinity to the target protein, representing an important advantage in medicinal chemistry.⁴⁷ Although benzylic ethers are less reactive to oxidative additions, Jarvo and co-workers revealed that 2-methoxyethyl ether could serve as an active substrate for C(sp³)–C(sp³) cross-coupling reactions (Scheme 18).⁴⁸ The treatment of 2-methoxyethyl

Scheme 18. Stereospecific Cross-Coupling of Benzylic 2-Methoxyethyl Ether with Methylmagnesium Iodide

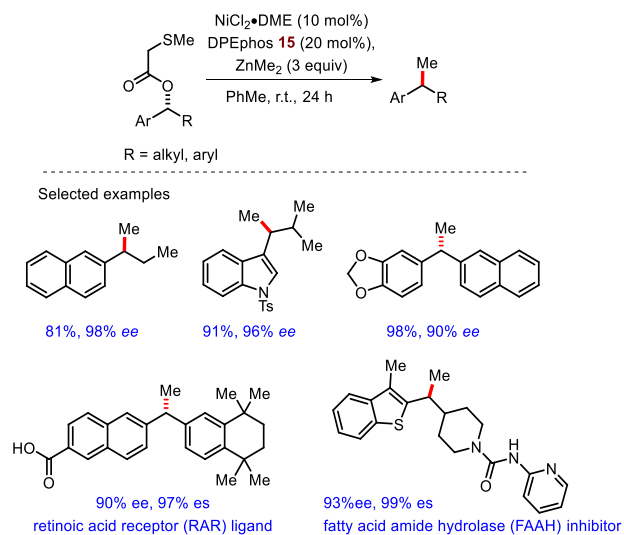


ethers with methylmagnesium iodide (2.0 equiv), DPEphos (20 mol %), and Ni(acac)₂ (10 mol %) in toluene at ambient temperature gave the corresponding coupled products. It was also found that the cross-coupling reaction with chiral substrates proceeds with an inversion of the configuration. The present method in which the oxidative addition was accelerated using methoxyethyl ether as a directing group delivered 1,1-diarylethanes and N-heterocycles with excellent enantiospecificity.

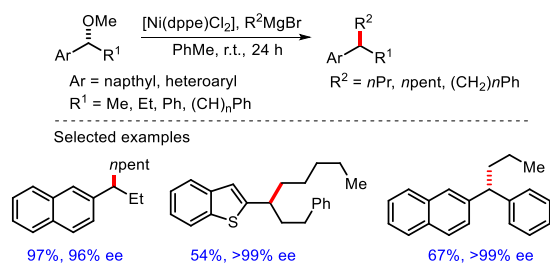
In many cases a Grignard reagent is used as a coupling partner to incorporate a methyl group. Inspired by the work of Liebeskind,⁴⁹ the Jarvo group disclosed a stereospecific Ni-catalyzed alkyl–alkyl cross-coupling strategy for the introduction of benzylic methyl substituents by introducing a thioester directing group to trigger the oxidative addition of a less reactive electrophile (Scheme 19).^{13b} The authors showcased that the installation of a 2-(methylthio)ester directing group at the benzylic position is an efficient approach to activate benzyl alcohols for the coupling reaction. The treatment of enantioriched thioesters with dimethylzinc reagent in the presence of NiCl₂·DME and DPEphos in toluene at room temperature afforded methylated products with a net inversion of the configuration (Scheme 19). Remarkably, this protocol tolerates a wide range of functional groups and a variety of heterocycles. The synthetic utility of this transformation was further demonstrated by the enantioselective synthesis of two bioactive molecules.

Subsequently, the Jarvo group in 2014 reported an enantioselective cross-coupling reaction of benzylic ethers with long-chain alkyl Grignard reagents (Scheme 20).^{13c} In the presence of 2 mol % of [Ni(dppe)Cl₂] and 2.0 equiv of Grignard reagent, the desired cross-coupling products could be isolated in high yields with good enantiospecificities. Further experiments demonstrated that aryl and heteroaryl Grignard reagents also serve as suitable substrates with this modified catalytic system. Interestingly, some of the resulting coupling

Scheme 19. Stereospecific Cross-Coupling of Secondary Benzylic Esters with Dimethylzinc



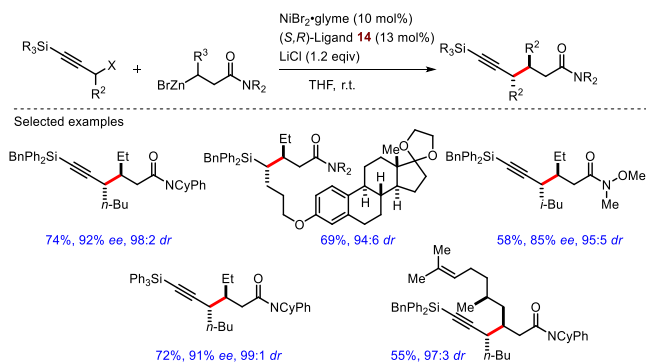
Scheme 20. Stereospecific Kumada Cross-Coupling of Chiral Benzylic Ethers



products exhibited a significant inhibition of breast cancer cell growth.

Another major advancement in alkyl–alkyl cross-coupling chemistry is Fu's development of double enantioconvergent coupling of racemic partners to control the vicinal stereocenters (Scheme 21).⁹

Scheme 21. Doubly Enantioconvergent Cross-Coupling of Propargylic Bromides with β -Zincated Amides



Building on their own early reports, Fu and co-workers further elaborated the alkyl–alkyl cross-coupling of racemic propargylic bromides with a variety of racemic β -zincated amides to control the vicinal stereocenters. The use of racemic nucleophiles in enantioconvergent substitution reactions remains a significant unexplored challenge. This work presented an efficient method

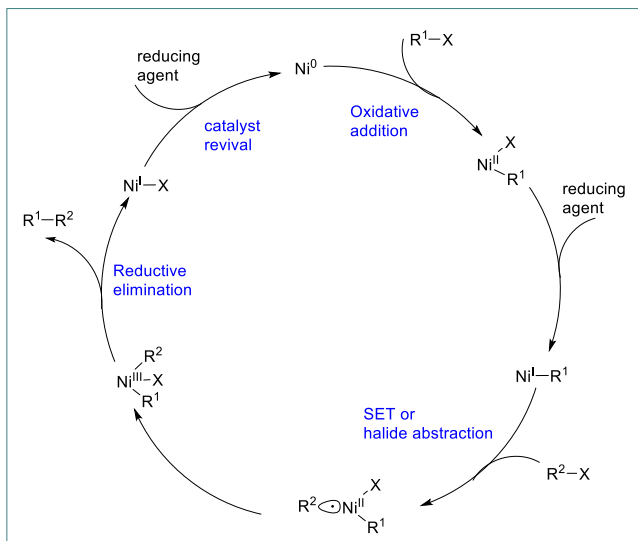
to realize the simultaneous control of stereochemistry at both carbons of the newly formed bond in superior yields. The optimized set of conditions involved catalytic $\text{NiCl}_2\cdot\text{glyme}$, an oxazolidine ligand, and 1.2 equiv of LiCl . The reaction was carried out in THF at room temperature for 20 h. A variety of functional groups such as an acetal, an ether, an ester, and silicon substituents were well tolerated under these mild reaction conditions. Although the reaction pathway is unknown, it is assumed that the mechanism involves the generation of radicals from both coupling partners, which then controls the double stereoconvergence.

CROSS-ELECTROPHILE COUPLING REACTIONS (XEC)

Cross-electrophile coupling reactions (XEC) have recently seen an increase in momentum.⁵⁰ Obtaining cross-selectivity can, of course, be a typical challenge, but the ready availability of two electrophilic coupling partners and the functional group compatibility is advantageous in XEC coupling reactions. Cross-selectivity mainly depends on the reactivity of alkyl halides;⁵¹ hence, most of these reactions are limited to the coupling between secondary and primary halides to avoid a competitive homocoupling. Due to the absence of an organometallic nucleophile, there is no transmetalation step in the cross-electrophile coupling reactions. Instead, it involves the two oxidative addition steps.⁵² An ample electronic differentiation between the two coupling partners is highly essential to favor cross-coupling over homocoupling because it allows one electrophile to be involved in oxidative addition with the $\text{Ni}(0)$ catalyst and the other to engage only in the second oxidative addition phase with the $\text{Ni}(I)$ intermediate.⁵⁰ The mechanisms that have been proposed so far for cross-electrophile coupling reactions have involved the alliance of sp^2 - and sp^3 hybridized electrophiles. However, a plausible mechanism for $\text{C}(\text{sp}^3)\text{--}\text{C}(\text{sp}^3)$ cross-electrophile coupling reactions proposed by Gong is depicted in Scheme 22.^{11a} Further mechanistic investigations are still desired to answer the remaining questions.

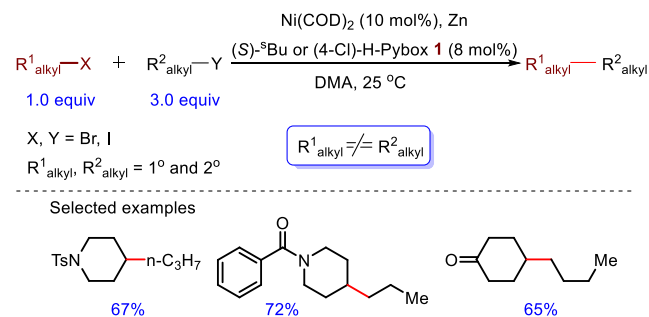
The Gong group were first to show the effective cross-coupling of two alkyl halides, introducing the nickel-catalyzed

Scheme 22. Possible Mechanism for $\text{C}(\text{sp}^3)\text{--}\text{C}(\text{sp}^3)$ Cross-Electrophile Coupling Reactions



reductive cross-coupling reaction as an alternative to the traditional cross-coupling strategy (Scheme 23).^{11a} The reaction

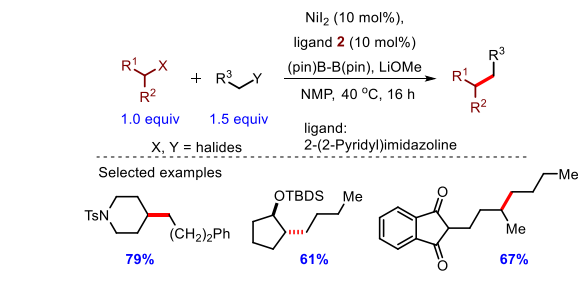
Scheme 23. Cross-Electrophile Coupling of Unactivated Alkyl Halides



was thought to proceed via the oxidative addition of low-valent Ni intermediates to both $\text{R}^1\text{--X}$ and $\text{R}^2\text{--X}$, which means that the two alkyl halides go through radical intermediates during the course of coupling; also, the possibility of organozinc/Negishi pathways was ruled out by carrying out control experiments. This method has the advantage of using alkyl halides that are more easily accessible and convenient to handle than the alkyl metallic nucleophiles.

To minimize the formation of homocoupling side products, the Gong group reported an extension of this cross-coupling reaction, by employing bis(pinacolato)diboron as the reductant (Scheme 24).^{11b} Mechanistic studies suggested that the *in situ*

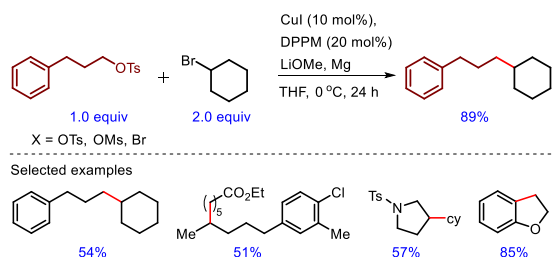
Scheme 24. Cross-Electrophile Coupling of Unactivated Alkyl Halides



formation of a alkyl-Bpin/Suzuki sequence was not operative. Instead, the reaction proceeded through the double oxidative addition of alkyl halides to Ni^I . Furthermore, the bases and ligands had a significant effect on the reaction outcome. Importantly, this present protocol was effective for coupling between secondary and primary alkyl halides but the reaction between two secondary alkyl halides was not very successful. Overall, this method addressed the key issues associated with poor selectivity (i.e., homocouplings) and use of excess amounts of one of the coupling partners.

Previously, the use of copper in reductive cross-coupling reaction between two unactivated alkyl halides was not known. In 2014, Liu and co-workers documented an efficient copper catalytic system for the reductive cross-coupling of alkyl-OTs/OMs with alkyl bromides (Scheme 25).⁵³ Interestingly, the bidentate phosphine ligand DPPM (bis(diphenylphosphino)methane) improved the product yields. However, the coupling between two alkyl halides furnished a significant amount of homocoupling product. The authors then extended the

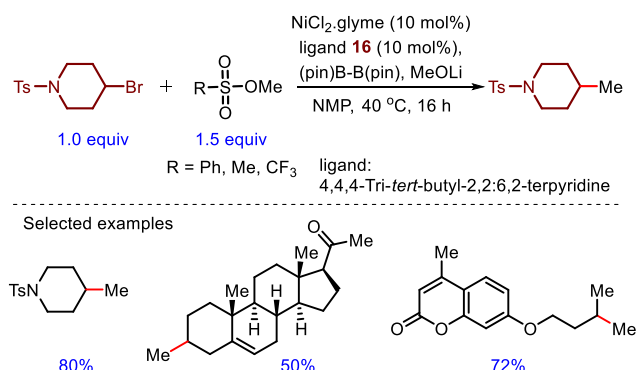
Scheme 25. Cross-Electrophile Coupling of Unactivated Alkyl Tosylates and Mesylates with Alkyl Bromides



procedure to intramolecular cyclization utilizing aryl bromides having an alkyl tosylate group chain.

As was mentioned earlier, installation of a methyl unit affords an increase in hydrophobic interactions, which results in an increase in the potency of the reaction.⁴⁷ Hence, this transformation is of course highly interesting in the context of medicinal chemistry and drug development. Several protocols have been reported for the methylation of aryl and alkyl carbons. In all of these cases, the use of Me-metallic nucleophiles was common. In an alternative approach, the Gong group disclosed a nickel-catalyzed reductive methylation of alkyl halides with methyl *p*-tosylate (Scheme 26).^{11c} It was found that methyl *p*-

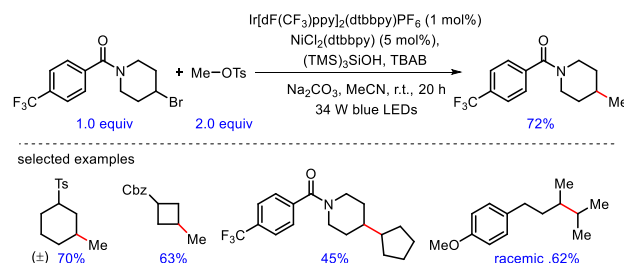
Scheme 26. Cross-Electrophile Coupling of Alkyl Halides with Methyl *p*-Tosylate



toluenesulfonates react with alkyl bromides in the presence of NiCl₂·glyme catalyst, a ligand, a (pin)B–B(pin) reductant, and MeOLi to give methylated alkanes. The authors believed that the reaction mechanism did not involve the formation of a methyl-Bpin intermediate.

The field of photocatalysis in C–C bond formation reactions has attracted significant interest, as it harnesses the energy of light and delivers it to a select substrate via a single-electron-transfer (SET) pathway.⁵⁴ In 2018, MacMillan and co-workers used metallaphotoredox catalysis for the cross-electrophile coupling of aliphatic bromides with the methyl tosylates (Scheme 27). Using 1 mol % of Ir[dF(CF₃)ppy]₂(dtbbpy)PF₆ as the photocatalyst, 5 mol % of (dtbbpy)Ni^{II}Cl₂, 1.5 equiv of tris(trimethylsilyl)silanol, and 2 equiv of Na₂CO₃ as a base, a substituted piperidine bromide was coupled with simple halides and methyl tosylates at room temperature.^{12a} The reaction was performed in MeCN for 20 h. It was observed that the use of tris(trimethylsilyl)silanol, a halide abstraction reagent, would be desirable in this reaction, as it forms silyl radicals by undergoing a photocatalytic silanolate oxidation and stops a hydrogen atom transfer with the alkyl radical intermediates, thereby preventing the generation of dehalogenated alkane products. The use of

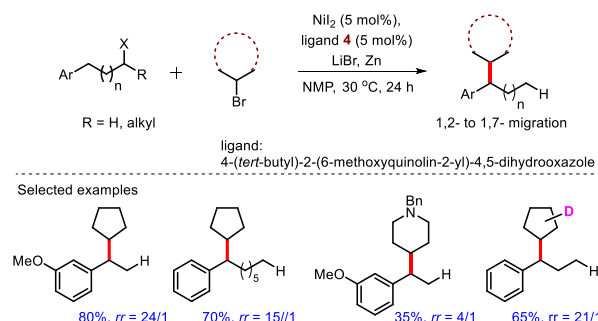
Scheme 27. Cross-Electrophile Coupling of Aliphatic Bromides



tetrabutylammonium bromide as an additive resulted in better yields.

In addition to the direct coupling methods cited above, the Zhu and Yin group was able to expand the scope by introducing an unprecedented chain walking of the metal center to facilitate the site-selective cross-coupling at positions away from the original metalation site.⁵⁵ In their study, primary alkyl bromides were treated with secondary alkyl bromides in the presence of the catalyst NiI₂ and a unique pyrOx ligand (Scheme 28). Under

Scheme 28. Migratory Alkyl–Alkyl Cross-Electrophile Coupling of Activated and Unactivated Alkyl Halides



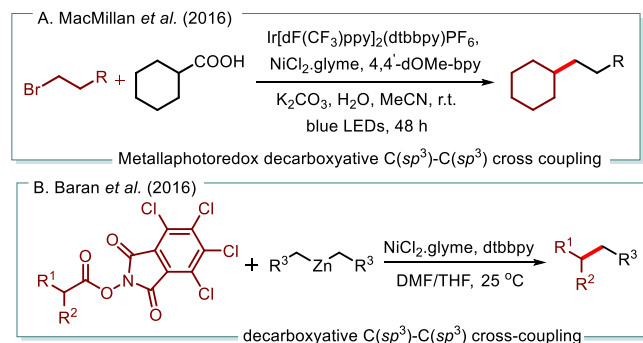
the optimized reaction conditions the nickel chain walking from the primary carbon to the benzylic carbon resulted in the benzylic alkylated products by installing a new 2°–2° carbon–carbon bond. The authors believed that the chain walking occurred at both coupling partners, and the conjecture was proven by deuterium migration. Interestingly, the use of the ligand 2,2′-bipyridine or 1,10-phenanthroline furnished the nonmigratory cross-coupled product and the exact reason for this outcome was unclear. Further study is desired to corroborate the reaction mechanism.

MISCELLANEOUS STRATEGIES

Carboxylic acids are chemically robust and inexpensive feedstocks for organic syntheses. The direct use of carboxylic acids in cross-coupling reactions has a long history.⁵⁶ It has been demonstrated that C(sp³) acids can serve as surrogates for the organometallic nucleophiles in metal-catalyzed cross-coupling reactions. However, the reports of decarboxylative C(sp³)–C(sp³) cross-coupling reactions are very rare. Further research is needed in this elegant area.

In 2016, MacMillan and co-workers made excellent use of metallaphotoredox catalysis for C(sp³)–C(sp³) cross-coupling (Scheme 29A).^{12b} The reaction involves single-electron oxidation of a carboxylate to provide a radical intermediate which then enters the catalytic cycle by intercepting an alkylnickel(II) complex. This strategy opens up a new avenue

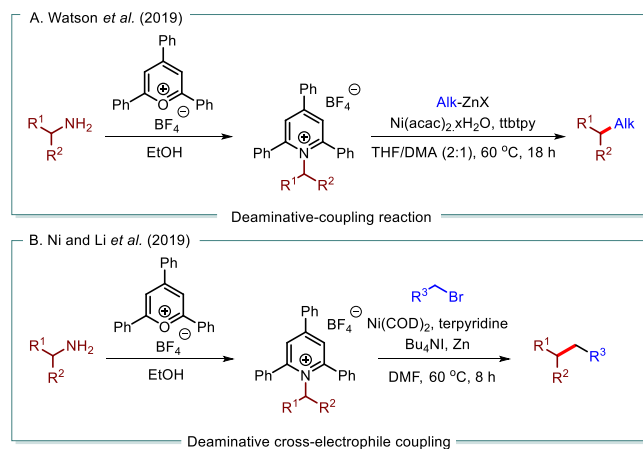
Scheme 29. Decarboxylative Cross-Coupling Reactions



for C(sp³)-C(sp³) cross-coupling reactions. Earlier work in generating alkyl radical intermediates through decarboxylation of carboxylic acid derivatives involved the use of photocatalysts.⁵⁷ In contrast, Baran and co-workers developed a decarboxylative C(sp³)-C(sp³) cross-coupling reaction without the aid of a photocatalyst.⁵⁸ The authors took the advantage of generating alkyl radicals from the *N*-hydroxyphthalimide (NHP) esters,⁵⁹ thereby forming new C(sp³)-C(sp³) bonds by reacting with dialkylzinc reagents using a low-valent nickel catalyst (Scheme 29B). This process also enables the smooth introduction of a conjugate acceptor, allowing for a three-component reaction that is beneficial for target-oriented synthesis.

The Negishi cross-coupling of alkylpyridinium salts with alkylzinc halides, reported by the Watson group in 2019, marked the first alkyl-alkyl bond formation via C-N bond activation (Scheme 30A).⁶⁰ The successful utilization of Katritzky salts as

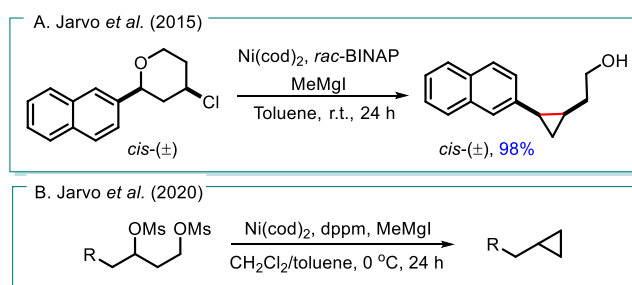
Scheme 30. Deaminative Cross-Coupling Reactions



carbon-centered radicals created an interest in this respective area. Similarly, the Ni and Li group developed a metal-catalyzed cross-electrophile coupling reaction of Katritzky salts with alkyl bromides (Scheme 30B).⁶¹

The intramolecular reductive cross-coupling of 2-aryl-4-chlorotetrahydropyrans to substituted cyclopropanes, reported by the Jarvo group in 2015, had a high stereospecificity, an unusual result.⁶² The reaction was efficiently catalyzed by Ni(cod)₂ and *rac*-BINAP, in the presence of MeMgI (Scheme 31A). Recent work from Jarvo, Hong, and co-workers described the Ni-catalyzed alkyl-alkyl cross-coupling reaction of 1,3-dimesylates for the synthesis of cyclopropanes (Scheme 31B).⁶³

Scheme 31. Intramolecular Reductive Cross-Electrophile Coupling Reactions



This transformation employs 5.0 mol % of Ni(cod)₂, 5.0 mol % of *rac*-BINAP ligand, and 2.0 equiv of methylmagnesium iodide (MeMgI) in DCM/PhMe.

SUMMARY AND OUTLOOK

In the past few years, the surge of advancements in transition-metal-catalyzed alkyl-alkyl cross-coupling reactions has allowed for the developments of versatile tools for C(sp³)-C(sp³) bond formation. In particular, nickel catalysts have played an essential role in forming a new carbon-carbon bond under milder conditions. Herein, an overview of advancements in traditional and XEC reaction manifolds for the construction of C(sp³)-C(sp³) bonds is provided. The strategies and methods showcased in this review demonstrated the power of improved catalytic systems for alkyl-alkyl bond formation. In addition, the collection of useful catalysts and ligands provide a powerful toolbox for the effective construction of bonds from a variety of coupling partners.

While the outstanding protocols establish a solid foundation for the area, a few difficulties have yet to be overcome. One such major challenge is the development of stereoselective approaches. To date, methods that control stereochemistry at both carbons of a new bond are uncommon. The difficulty in stereocontrol via conventional two-electron migratory insertion processes might be one of the reasons for this challenge. Future developments using one-electron, ligand-based redox concepts may address challenging problems and provide advanced asymmetric versions using chiral ligands. The enantioselectivity of the coupled product also correlates with the free energy differences between two transition states of the reductive elimination step. In this context, methods that calculate the relative rates of product formation in both *R* and *S* enantiomers will hopefully benefit researchers in the design of effective catalytic systems and coupling partners for asymmetric synthesis. In addition, reducing the high catalyst loadings and developing low-cost ligands and robust nickel systems are much desired for industrial applications.

Further detailed mechanistic investigations of these transformations will provide insight into the rational design and development of metal-catalyzed cross-coupling reactions. The potential scope for future expansion involves the introduction of novel catalytic systems for the stereoselective cross-coupling reactions of racemic alkyl coupling partners. The advent of these protocols would provide a roadmap for strategic disconnections to simplify the synthesis of chiral frameworks. Doubtless, as it develops, the asymmetric cross-coupling reaction will become more and more general and be used for challenging bond formation in the synthesis of natural products, pharmaceuticals, and other more complex molecules.

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Notes

The author declares no competing financial interest.

Biography



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